

```
In [1]: # Cài đặt các thư viện cần thiết
!pip install gensim matplotlib scikit-learn numpy
```

```
Collecting gensim
  Obtaining dependency information for gensim from https://files.pythonhosted.org/packages/10/c3/7e22d6f7d88c4ea6a3a84481f00538252659d285713c3b7e2e1537b0e7e1/gensim-4.4.0-cp311-cp311-win_amd64.whl.metadata
  Using cached gensim-4.4.0-cp311-cp311-win_amd64.whl.metadata (8.6 kB)
Requirement already satisfied: matplotlib in c:\users\hoang\appdata\local\programs\python\python311\lib\site-packages (3.10.7)
Collecting scikit-learn
  Obtaining dependency information for scikit-learn from https://files.pythonhosted.org/packages/9f/71/34ddbd21f1da67c7a768146968b4d0220ee6831e4bcbad3e03dd3eae88b6/scikit_learn-1.7.2-cp311-cp311-win_amd64.whl.metadata
  Using cached scikit_learn-1.7.2-cp311-cp311-win_amd64.whl.metadata (11 kB)
Requirement already satisfied: numpy in c:\users\hoang\appdata\local\programs\python\python311\lib\site-packages (1.26.4)
Collecting scipy>=1.7.0 (from gensim)
  Obtaining dependency information for scipy>=1.7.0 from https://files.pythonhosted.org/packages/f1/d0/22ec7036ba0b0a35bccb7f25ab407382ed34af0b111475eb301c16f8a2e5/scipy-1.16.3-cp311-cp311-win_amd64.whl.metadata
  Using cached scipy-1.16.3-cp311-cp311-win_amd64.whl.metadata (60 kB)
Collecting smart_open>=1.8.1 (from gensim)
  Obtaining dependency information for smart_open>=1.8.1 from https://files.pythonhosted.org/packages/ad/95/bc978be7ea0babf2fb48a414b6afaad414c6a9e8b1eafc5b8a53c030381a/smart_open-7.5.0-py3-none-any.whl.metadata
  Using cached smart_open-7.5.0-py3-none-any.whl.metadata (24 kB)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\hoang\appdata\local\programs\python\python311\lib\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\users\hoang\appdata\local\programs\python\python311\lib\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\hoang\appdata\local\programs\python\python311\lib\site-packages (from matplotlib) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\hoang\appdata\local\programs\python\python311\lib\site-packages (from matplotlib) (1.4.9)
Requirement already satisfied: packaging>=20.0 in c:\users\hoang\appdata\local\programs\python\python311\lib\site-packages (from matplotlib) (23.2)
Requirement already satisfied: pillow>=8 in c:\users\hoang\appdata\local\programs\python\python311\lib\site-packages (from matplotlib) (12.0.0)
Requirement already satisfied: pyparsing>=3 in c:\users\hoang\appdata\local\programs\python\python311\lib\site-packages (from matplotlib) (3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\hoang\appdata\local\programs\python\python311\lib\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: joblib>=1.2.0 in c:\users\hoang\appdata\local\programs\python\python311\lib\site-packages (from scikit-learn) (1.5.2)
Collecting threadpoolctl>=3.1.0 (from scikit-learn)
  Obtaining dependency information for threadpoolctl>=3.1.0 from https://files.pythonhosted.org/packages/32/d5/f9a850d79b0851d1d4ef6456097579a9005b31fea68726a4ae5f2d82ddd9/threadpoolctl-3.6.0-py3-none-any.whl.metadata
  Using cached threadpoolctl-3.6.0-py3-none-any.whl.metadata (13 kB)
Requirement already satisfied: six>=1.5 in c:\users\hoang\appdata\local\programs\python\python311\lib\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
Requirement already satisfied: wrapt in c:\users\hoang\appdata\local\programs\python\python311\lib\site-packages (from smart_open>=1.8.1->gensim) (1.17.3)
Using cached gensim-4.4.0-cp311-cp311-win_amd64.whl (24.4 MB)
Using cached scikit_learn-1.7.2-cp311-cp311-win_amd64.whl (8.9 MB)
Using cached scipy-1.16.3-cp311-cp311-win_amd64.whl (38.7 MB)
Using cached smart_open-7.5.0-py3-none-any.whl (63 kB)
Using cached threadpoolctl-3.6.0-py3-none-any.whl (18 kB)
Installing collected packages: threadpoolctl, smart_open, scipy, scikit-learn, gensim
Successfully installed gensim-4.4.0 scikit-learn-1.7.2 scipy-1.16.3 smart_open-7.5.0 threadpoolctl-3.6.0
[notice] A new release of pip is available: 23.2.1 -> 25.3
[notice] To update, run: python.exe -m pip install --upgrade pip
```

```
In [2]: import numpy as np
import matplotlib.pyplot as plt
from sklearn.decomposition import PCA
from sklearn.manifold import TSNE
import gensim.downloader as api

# Load model pre-trained (Glove 50d)
# Model này nhẹ (65MB), phù hợp để demo trên Colab
print("Đang tải model...")
model = api.load('glove-wiki-gigaword-50')
print("Tải model thành công!")
```

```
Đang tải model...
[=====] 100.0% 66.0/66.0MB downloaded
Tải model thành công!
```

```
In [8]: # Danh sách các từ để trực quan hóa
word_groups = {
    'Family': ['father', 'mother', 'brother', 'sister', 'grandfather', 'grandmother', 'son', 'daughter'],
    'Colors': ['red', 'blue', 'green', 'yellow', 'orange', 'purple', 'white', 'black'],
    'Technology': ['computer', 'keyboard', 'mouse', 'screen', 'software', 'hardware', 'internet', 'wifi'],
    'Animals': ['dog', 'cat', 'lion', 'tiger', 'elephant', 'monkey', 'bird', 'fish']
}
```

```

# Gom tất cả từ lại và lấy vector tương ứng
all_words = []
all_vectors = []
labels = []

for group, words in word_groups.items():
    for w in words:
        if w in model:
            all_words.append(w)
            all_vectors.append(model[w])
            labels.append(group) # Lưu nhãn nhóm để tô màu

all_vectors = np.array(all_vectors)
print(f"Số lượng từ: {len(all_words)}")
print(f"Kích thước vector gốc: {all_vectors.shape}")

```

Số lượng từ: 32

Kích thước vector gốc: (32, 50)

```

In [9]: def plot_embeddings(vectors_2d, words, labels, title):
    plt.figure(figsize=(12, 8))

    # Tạo màu cho từng nhóm
    unique_labels = list(set(labels))
    colors = plt.cm.rainbow(np.linspace(0, 1, len(unique_labels)))

    for i, group_label in enumerate(unique_labels):
        # Lấy index của các từ thuộc nhóm này
        idxs = [j for j, label in enumerate(labels) if label == group_label]

        # Vẽ điểm scatter
        plt.scatter(vectors_2d[idxs, 0], vectors_2d[idxs, 1],
                    c=[colors[i]], label=group_label, s=100, alpha=0.7)

        # Gán nhãn tên từ lên biểu đồ
        for idx in idxs:
            plt.annotate(words[idx],
                        xy=(vectors_2d[idx, 0], vectors_2d[idx, 1]),
                        xytext=(5, 2),
                        textcoords='offset points',
                        ha='right',
                        va='bottom')

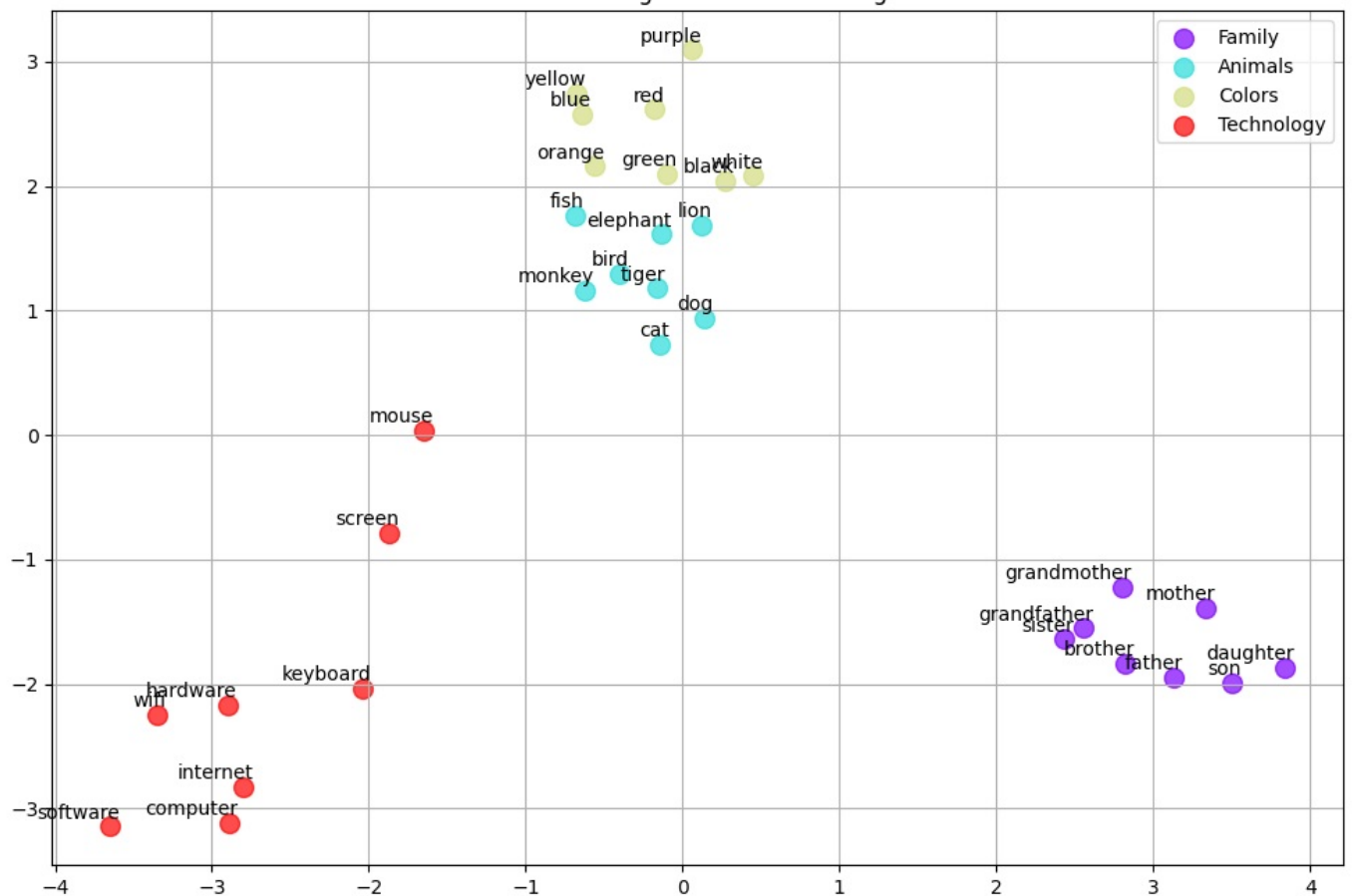
    plt.legend(loc='best')
    plt.title(title)
    plt.grid(True)
    plt.show()

# --- Thực hiện PCA ---
pca = PCA(n_components=2)
vectors_pca = pca.fit_transform(all_vectors)

# Vẽ biểu đồ
plot_embeddings(vectors_pca, all_words, labels, "Word Embeddings Visualization using PCA")

```

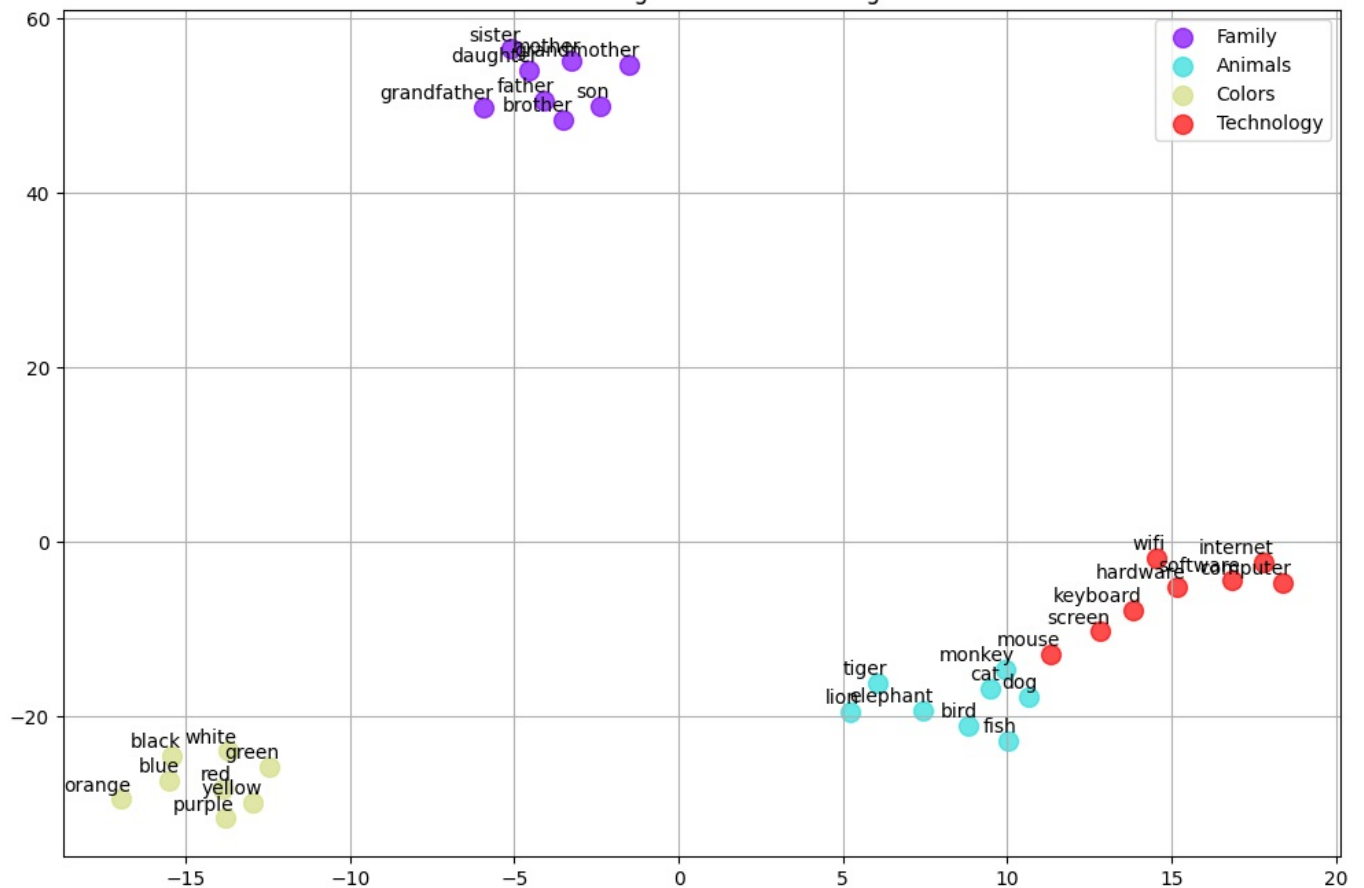
Word Embeddings Visualization using PCA



```
In [5]: # Perplexity nên nhỏ hơn số lượng điểm dữ liệu (thường từ 5-50).
# Vì tập data demo nhỏ (32 từ), ta chọn perplexity thấp (ví dụ: 5 hoặc 10).
tsne = TSNE(n_components=2, perplexity=5, random_state=42, init='pca', learning_rate='auto')
vectors_tsne = tsne.fit_transform(all_vectors)

# Vẽ biểu đồ
plot_embeddings(vectors_tsne, all_words, labels, "Word Embeddings Visualization using t-SNE")
```

Word Embeddings Visualization using t-SNE



In []:

